Kyle A. Moore

moore.kyle.andrew@gmail.com | kyle.a.moore@vanderbilt.edu | (901)734-6720
GitHub: KyleAMoore | LinkedIn: kyle-andrew-moore | Website: kyleamoore.github.io

RESEARCH INTERESTS

Artificial Intelligence & Machine Learning · Natural Language Processing · Large Language Models · AI Behavior & Human–AI Comparison · Human-AI Interaction (future direction) · Affective Computing (future direction)

EDUCATION

Ph.D., Computer Science – Vanderbilt University *Aug 2019 – Sep 2026 (expected)*
Dissertation: "Human-Analogous Uncertainty Behavior: Cognitive Studies of Large Language Models"
Advisor: Doug Fisher

M.S., Computer Science – University of Mississippi *Aug 2019 – May 2019*
Thesis: "Building an Automated QA System Using Online Forums as Knowledge Bases "
Advisor: Naeemul Hassan

B.S., Computer Science – University of Mississippi *Aug 2013 – May 2017*

PUBLICATIONS

Peer-Reviewed

- Moore, K., Roberts, J., Watson, D., Wisniewski, P. "Investigating Human-Aligned Large Language Model Uncertainty." FLAIRS 2026.
- Sawyer, H., Roberts, J., Moore, K. "Basic Categories in Vision Language Models: Expert Prompting Doesn't Grant Expertise." ACS 2025.
- Moore, K., Roberts, J., Pham, T., Fisher, D. "Chain of Thought Still Thinks Fast: APriCoT Helps with Thinking Slow." CogSci 47, 2025.
- Moore, K., Roberts, J., Pham, T., Ewaleifoh, O., Fisher, D. "The Base-Rate Effect on LLM Benchmark Performance: Disambiguating Test-Taking Strategies from Benchmark Performance." EMNLP 2024.
- Roberts, J., Moore, K., Fisher, D., Ewaleifoh, O., Pham, T. "Large Language Model Recall Uncertainty is Modulated by the Fan Effect." CoNLL 28, 2024.
- Roberts, J., Moore, K., Wilenzick, D., Fisher, D. "Using Artificial Populations to Study Psychological Phenomena in Neural Models." AAAI 38, 2024.
- Timalsina, U., Broll, B., Moore, K., Lédeczi, Á. "DeepForge for Astronomy: Deep Learning SDSS Redshifts from Images." Astronomy and Computing 40 (2022).

Workshop Papers

- Roberts, J., Moore, K., Fisher, D. "Do Large Language Models Learn Human-Like Strategic Preferences?" REALM Workshop, ACL 2025.

Preprints/Under Review

- Moore, K., Roberts, J., Watson, D., Ward, W., Heyboer, G. "Human-Alignment, Calibration, and Activation Patterns in Large Language Model Uncertainty." Preprint, under review, 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant – Vanderbilt University Aug 2019 – Present

- Designed and ran controlled behavioral studies testing whether LLMs exhibit human-analogous patterns of uncertainty behaviors like typicality and fan effects, bridging cognitive science methodology and NLP evaluation.
- Showed that answer-token base rates, not knowledge, drive much of reported MMLU accuracy across 5 open LLMs; introduced Nvr-X, a dataset-agnostic augmentation procedure that disentangles task performance from test-taking strategy in MCQA benchmarks, under which only 1 of 5 models beat random chance.
- Established the first direct human-vs-LLM uncertainty comparison on paired tasks, scaling to 30 open-weight models and 4 QA datasets; found that instruct fine-tuning degrades both human-alignment and calibration, and that internal activation probes recover substantially more alignment signal than output logits.

- Mentored 6 undergraduate researchers from problem formulation through publication, co-authoring 6 papers.

TEACHING EXPERIENCE

Instructor - CS 4260: Introduction to Artificial Intelligence – Vanderbilt University Spring 2025

- Designed the course and taught it as sole lecturer to ~60 undergraduate and 2 graduate students, with full responsibility for syllabus, materials, assessment, and grading. Graduate students are not ordinarily appointed to teach at Vanderbilt; the arrangement was documented by a letter from the Dean. Course listed under faculty advisor Doug Fisher, who held the formal appointment. More details, including said letter, are available on request.
- Covered topics include: goal-directed search, adversarial search, automated theorem proving, logic-based planning, CSPs, decision trees, neural networks, clustering, Bayesian probability, belief nets, (hierarchical hidden) Markov models, MDPs, RL

Teaching Assistanceships – Ph.D. – Vanderbilt University (* - guest lectures given)

- CS 4260: Artificial Intelligence (2022F, 2023F, 2024S*, 2025F*, 2026S*)
- CS 1151: Computer Ethics (Summer 2023, Summer 2024, Summer 2025)
- CS 3252: Theory of Automata, Formal Languages, and Computation (2021F, 2023S*, 2024F*)
- CS 3265: Databases (Summer 2022, Summer 2023)
- CS 3891: Computational Creativity (2023S*)
- CS 3892: Software for Autonomous Vehicles (2022S)
- CS 3270: Programming Languages (2019F, 2020S, 2020F. 2021S)

RESEARCH PRESENTATIONS

- Poster, AAAI 2024
- Poster, EMNLP 2024
- Poster, CogSci 2025
- Talk, FLAIRS 2026

PROFESSIONAL SERVICE

Peer Review

- ICLR 2026
- EMNLP 2026
- FLAIRS 2026
- AAAI 2026
- ACS 2025
- ACL Rolling Review (ARR), February 2025
- REALM Workshop, ACL 2025

Organizing Committee

- FLAIRS 2026 Special Track: Human-AI Collaboration and Augmented Intelligence (HAICAI)

Other Scholarly Contributions

- Acknowledged for pre-publication feedback on Anderson & Fisher, Artificial Intelligence for Academic Libraries (2025 Routledge)

AWARDS & HONORS

- IBM Goldstine Graduate Fellowship, Vanderbilt University

MENTORING & ADVISING

Undergraduate Research Intern Mentees (includes only mentees who earned authorship on a resulting paper, published or under review):

- Drew Wilenzick (AAAI 2024)
- Oseremhen Ewaleifoh (CoNLL 2024; EMNLP 2024)
- Thao Pham (CoNLL 2024; EMNLP 2024; CogSci 2025)
- Daryl Watson (FLAIRS 2026; preprint 2026)
- William Ward (preprint 2026)

- Greyson Heyboer (preprint 2026)

TECHNICAL SKILLS

Languages: Python (primary), Java, C/C++, SQL, Haskell, LaTeX
ML/Data: NumPy, pandas, Matplotlib, TensorFlow, PyTorch, scikit-learn, HF transformers
Methods: uncertainty quantification and calibration, activation probing, nonparametric statistics
(Natural) Languages: Japanese (Elementary Proficiency)

GRADUATE COURSES TAKEN

Ph.D. – Vanderbilt University

- CS 8395 – Special Topics: Computational Game Theory (2021F)
- CS 6388 – Model-Integrated Computing (2020F)
- CS 5891 – Special Topics: Social Network Analysis (2020F)
- CS 8395 - Special Topics: Visual Analytics and Machine Learning (2020S)
- CS 6360 – Advanced Artificial Intelligence (2020S)
- EECE 6354 – Advanced Real-Time Systems (2019F)
- CS 8395 – Special Topics: Computation and Cognition (2019F)
- CS 6376 – Hybrid and Embedded Systems (2019F)

M.S. – University of Mississippi

- CSCI 556 – Multiparadigm Programming (2018F)
- CSCI 561 – Computer Networks (2018F)
- CSCI 547 – Digital Image Processing (2018F)
- CSCI 581 – Special Topics: Compiler Design (2018S)
- CSCI 581 - Special Topics: Natural Language Processing (2018S)
- CSCI 531 - Artificial Intelligence (2018S)
- ENGR 692 – Special Topics: High-Performance Computing (2017F)
- ENGR 691 – Special Topics: Random Algorithms (2017F)
- CSCI 533 – Analysis of Algorithms (2017F)